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Analysis of hydraulic characteristics of submerged gate discharge based on hybrid artificial intelligence models

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ABSTRACT

With the expansion of water conservancy projects and the complexity of water resources scheduling problems, calculating submerged flow has become increasingly critical and challenging, and improving its computational efficiency and accuracy has become a key problem. This study analyzes the hydraulic characteristics of sluice gate submerged flow based on a long-distance water transmission project. Random Forest (RF) and its optimization algorithm Firefly Algorithm-optimized Random Forest (FA-RF) were used to build an intelligent prediction model, with performance evaluated through index and uncertainty analysis. The Sobol sensitivity analysis method is used to quantify the hydraulic parameters affecting submerged flow, identifying core parameters influencing the submerged discharge coefficient C_d. The results show that for the submerged flow, the greater downstream water depth causes stronger upstream flow disturbance. In the testing phase, the indicators of the FA-RF model are better than RF, and the mean absolute percentage error (MAPE) and root mean square error (RMSE) are reduced by 20.9 % and 24.3 %, respectively, indicating that the firefly algorithm optimizes the performance of the RF algorithm. The upstream Froude number *Fr* and the ratio of downstream water depth to gate opening y_3/e were identified as core parameters affecting C_d. C_d increases with *Fr* and decreases with y_3/e.

1. Introduction

With rapid urbanization and increasing extreme weather events due to climate change, water resource management, particularly flood control and detention, has grown increasingly complex. A critical technical challenge is the rapid and accurate prediction of gate operation impacts on inundation areas and downstream discharge. Submerged flow calculation, as a key technology, holds significant value in flood management, environmental protection and optimal utilization of water resources. The complexity of the river system and expanding hydraulic infrastructure have rendered these calculations more challenging. Traditional physical models, while addressing complex topography, transient flows, and dynamic boundary conditions, often demand substantial computational resources and exhibit low efficiency. Thus, enhancing computational accuracy and efficiency has become an urgent problem.

Traditional hydrodynamic and physical models are widely used for submerged flow prediction in practical engineering. Hashem T et al. [1] conducted experiments using a rectangular channel to study the hydraulic characteristics and energy dissipation of a labyrinth sluice gate. The experiment emphasized the role of gate opening size in the hydraulic jump formation and its associated energy dissipation. Maghsoodi R et al. [2] used the computational fluid dynamics (CFD) software Fluent to simulate the submerged weir flow and analyze the flow field characteristics, demonstrating high accuracy consistent with physical hydraulic model data. Gumus V et al. [3] investigated the hydraulic jump of submerged flow of a sluice gate by experiment and numerical simulation, finding that the Reynolds stress model can better predict the horizontal velocity and surface profile, especially in the transition region. Ma F et al. [4] selected the standard k-ε turbulence model to analyze the hydraulic characteristics of submerged hydraulic jumps, including flow patterns, velocity, pressure and Froude number. Ferro et al. [5] deduced the dimensionless relationship of the orifice discharge of the sluice gate by using the π theorem of dimensional analysis and the incomplete self-similarity concept (ISS) theory. Shahrokhnia et al. [6] obtained the dimensionless relationship between the free and submerged flow of the radial gate by the dimensional analysis method. Bijankhan M et al. [7] improved the original dimensionless formula proposed by Ferro, proposing a more accurate equation applicable from free to submerged flow conditions. Daneshfaraz R et al. [8] studied gate

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hydraulic characteristics and established nonlinear regression polynomial relationships to predict energy dissipation and discharge coefficients. These methods are based on the CFD principle, simulating the dynamic evolution and submerged expansion of water flow. However, there are obvious limitations in the calculation process. First, the traditional method requires high simulation accuracy for complex terrain and large-scale water areas, as well as high-frequency time stepping and detailed meshing, resulting in substantial computational overhead. Additionally, these models poorly adapt to non-uniform boundary conditions and sudden floods, making them inefficient for real-time water resource scheduling and dynamic flood emergency management.

With the improvement of computing power, machine learning technology, as an emerging intelligent computing tool, has shown great potential in submerged flow prediction. By learning historical data, machine learning algorithms can quickly predict changes in water flow and submerged areas without relying on precise physical formulas. Roushangar K et al. [9] offered grey wolf optimization-based KELM (GWO-KELM), support vector machine (SVM) and Gaussian process regression (GPR) methods to predict the discharge coefficient of the submerged radial gates. The results reflected that GWO-KELM with RBF kernel function gives better prediction accuracy. Salmasi F et al. [10] used genetic programming (GP) to predict the discharge coefficient of free and submerged inclined sliding gates, showing greater efficiency than traditional regression models. Tao H et al. [11] developed Gaussian process regression (GPR), generalized regression neural network (GRNN), and multi-gene genetic programming (MGGP) to predict the discharge coefficient of radial gates under free and submerged conditions. Correlation analysis and sensitivity analysis were carried out, and the key variables affecting the flow coefficient were determined. Hashem T et al. [12] applied senven machine learning algorithms, including gradient Boosting model GBM, KNeighbors Regression(KNN), Huber Regression (HR), Sup-port Vector Regression (SVR) Linear and Radial Basis Functions (RBF), Random Forest (RF), and Linear Regression (LR), to estimate the discharge coefficient of labyrinth sluice gates. The study found the high predictive accuracy of the GB model compared to traditional methods. Hashem T et al. [13] employed regression techniques, Gene Expression Programming (GEP), KStar, and M5Prime (M5P), to investigate the discharge coefficient of labyrinth sluice gates. Among these, GEP showed the best performance, and the water depth contraction ratio and internal angle are the most significant factors influencing discharge coefficient. Machine learning models trained on historical hydrological data can rapidly predict submerged flow changes, demonstrating strong adaptability to complex terrain, climate variability, and sudden flow events. However, these models face significant limitations, primarily their inherent “black-box” nature, which results in a lack of transparency and physical interpretability [14]. In high-precision hydraulic engineering applications, such as safety assessments and early warning systems, the absence of clear physical mechanisms may hinder their reliability and restrict their broader adoption.

Therefore, enhancing the interpretability of machine learning models has become a research focus, aiming to shift towards more transparent AI [15]. For high-precision prediction requirements, higher interpretability can enhance transparency and engineering adaptability, enabling researchers to better understand model performance and decision-making processes under diverse hydrological conditions [16]. Especially in calculating submerged flow with multi-physical coupling and nonlinear characteristics, the interpretability can effectively bridge the gap between the machine learning and the traditional physical models, and ensure alignment with actual project requirements. Consequently, this study employed numerical simulation to analyze the hydraulic characteristics of the three-dimensional flow field of the submerged sluice gate, and selected Random Forest (RF) and its optimization algorithm Firefly Algorithm-optimized Random Forest (FA-RF) to build the intelligent model for submerged flow prediction. By conducting interpretability analysis of hydraulic parameters, selecting core parameters, and analyzing their variation patterns, the study aims to build a high-precision, interpretable hybrid model. This model reveals the mechanism of gate submerged flow, enriches systematic research on submerged flow calculation, and provides a theoretical basis for water resources scheduling and flood management.

2. Data and methods

2.1. Experimental setup

2.1.1. Numerical model

When the channel flows under the gate, the area of the section decreases sharply, leading to a dramatic change in flow. The flow presents prominent three-dimensional characteristics. FLOW-3D bases its strategy to model and track the free surface on the volume of fluid (VOF) method [17]. The RNG k-ε turbulence adds an extra source term and functions to the equations, performs better in dealing with rapidly strained and swirling flows [18], and is usually the best choice.

RNG k-ε model is a two-equation model, including turbulent kinetic energy (k) and turbulent dissipation rate(ε) [19].

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left[\left(\alpha_k \mu_{\varepsilon ff}\right)\frac{\partial k}{\partial x_i}\right]+G_k-\rho\varepsilon \tag{1}$$

$$\frac{\partial}{\partial t}(\rho \varepsilon)+\frac{\partial}{\partial x_i}(\rho \varepsilon u_i)=\frac{\partial}{\partial x_j}\left[\left(\alpha_\varepsilon \mu_{\varepsilon ff}\right)\frac{\partial \varepsilon}{\partial x_i}\right]+G_{1\varepsilon}\frac{\varepsilon}{k}G_k+G^*_{2\varepsilon}\rho\frac{\varepsilon^2}{k} \tag{2}$$

where $\mu_{\varepsilon ff}$ is the effective viscosity.

$$\mu_{eff}=\mu+\mu_t \tag{3}$$

$$\mu_t=\rho C_\mu \frac{k^2}{\varepsilon} \tag{4}$$

$$C^*_{1\varepsilon}=C_{1\varepsilon}-\frac{\eta(1-\eta/\eta_0)}{1+\beta\eta^3} \tag{5}$$

$$\eta=\left(2E_{ij}\cdot E_{ij}\right)^{1/2}\frac{k}{\varepsilon} \tag{6}$$

$$E_{ij}=\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right) \tag{7}$$

The constant values obtained from the RNG theory are: $G_{1\varepsilon}=1.42$, $G_{2\varepsilon}=1.68$, $G_\mu=0.0845$, $\alpha_k=\alpha_\varepsilon=1.393$, $\eta_0=4.38$, $\beta=0.012$ [20].

Near-wall turbulence was treated using FLOW-3D’s automatic wall treatment, which adaptively applies wall functions or resolves the viscous sublayer according to local grid spacing. The meshensured $1\leq y^+\leq 5$ near solid boundaries. Time-step independence was confirmed by halving $\triangle$t, with variations in C_d, and free-surface elevation below 1 %, demonstrating numericaconvergence [21].

2.1.2. Geometric model

In a long-distance water conveyance project, the length of the south trunk line is 103.10 km, and the design discharge at the beginning is 47 m^3/s. The axis of the inlet gate of the south trunk line is 0 + 031 m on the right side of the diversion pool, with a length of 15.6 m. The channel includes a culvert section and a gate chamber section. The culvert is connected with the diversion pool, the length is 6.6 m, the size of the concrete culvert is 5 × 6.5 m, and the bottom elevation is 510 m. The length of the gate chamber is 9m, a single hole with a width of 5m, and the bottom elevation of the inlet gate is 510 m. Two gates with a size of 6 × 5 m are installed in the gate chamber. The gate chamber is connected to the south trunk tunnel. The tunnel’s cross-section is a horseshoe shape with a size of 5.97 × 5.58 m. The longitudinal gradient of the tunnel is 1/2500, and the design water elevation of the diversion pool is 514.88 m, 513.08 m, and 512.00 m.

This study established a three-dimensional geometric model of the diversion pool and the south trunk channel based on the project. As shown in Fig. 1, the length of the diversion pool is 52.5 m, and the width is 35 m. The channel cross-section at the entrance of the south trunk line is 5 × 6.5 m, and the longitudinal ratio is 1/2500. A transition section totaling 19 m was set up at 6m downstream of the channel inlet. The cross-section gradually changes from a rectangular shape to a city-gate shape to a horseshoe shape. The height of the city-gate-shaped cross-section is 6.2 m, and the width is 5 m. The height of the horseshoe-shaped section is 5.58 m, and the width is 5.97 m.

2.1.3. Computational mesh

Reasonable mesh size and boundary conditions determine whether the computational solution is efficient. The Grid Convergence Index (GCI) [22] was used to calculate the grid convergence for the sluice gate structure, which is expressed for the fine grid solution as:

$$GCI_{i+1,i} = \frac{F_s}{r^p - 1} \cdot re_{i+1,i} \tag{8}$$

$$re_{i+1,i} = \left| \frac{f_{i+1} - f_i}{f_i} \right| \tag{9}$$

$$r = \frac{D_i}{D_{i+1}} \tag{10}$$

where F_s is a safety factor, and its range is 1.25–3.0. There are three different grid spacings in this paper, so $F_s = 1.25$; p is the convergence accuracy; $re_{i+1,i}$ is relative error; f_i is the grid solutions for the fine grid resolution; r is the grid refinement ratio; D denotes the mesh size [23].

As shown in Table 1, all GCI values are below 5 %, indicating that the results have reached satisfactory convergence. When the grid size is less than 1 cm, the GCI value diminishes to below 1 %, and the result is not dependent on the grid size. However, D = 1 cm substantially increases the number of computational cells and simulation time, resulting in a disproportionately high computational cost. Considering the balance between computational efficiency and accuracy, D = 2 cm is selected as the optimal mesh size(see Fig. 2). The total number of grid cells is approximately 2,830,000, ensuring a balance between accuracy and computational expense.

2.1.4. Boundary conditions

The simulation of the submerged flow was conducted in FLOW-3D using the air-water VOF model. The initial conditions for simulating submerged flow were the upstream and downstream water elevations. Boundary condition setting: the X_{min} direction of block 1 is the pressure inlet, the X_{max}, Y_{min}, Y_{max} and Z_{min} directions are Walls, and Z_{max} is the air inlet. Mesh block 2 includes the channel and the sluice gate, the X_{max}, X_{min}, Y_{min}, Y_{max} and Z_{min} are Walls, and Z_{max} is the air inlet. The Y_{min} direction of block 3 is the pressure outlet, the X_{max}, X_{min}, Y_{max}, and Z_{min} are Walls, and Z_{max} is the air inlet. The overall boundary conditions are set as presented in Fig. 3.

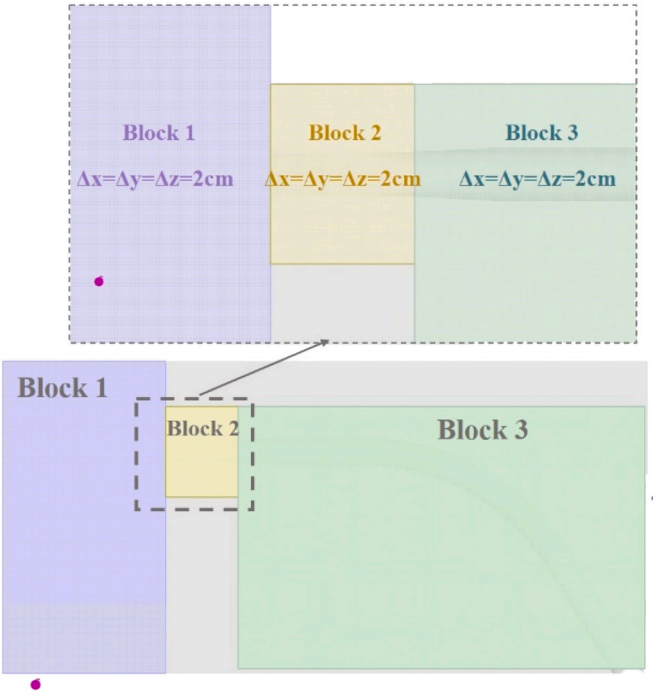


Fig. 2. Meshing for submerged flow simulation of sluice gate.

Table 1
Mesh independence analysis using the GCI method.

D	F_s	p	r (D_i/D_{i+1})	Relative error (re)	GCI(%)
3	1.25	1.75	–	–	–
2	1.25	1.75	1.5	0.03846	3.15
1	1.25	1.75	2	0.02767	0.86

2.1.5. Condition setting

The simulation condition setting is shown in Table 2. The initial conditions include upstream elevation, downstream water depth and gate opening. The initial upstream water elevation is 513.896 m, 513.08 m and 512.00 m, respectively. Each elevation corresponds to 20 gate openings and five downstream water depths. In this study, 300 simulated datasets were implemented to feed the employed ML methods.

2.1.6. Model validation

To verify the reliability of the numerical simulation results, the

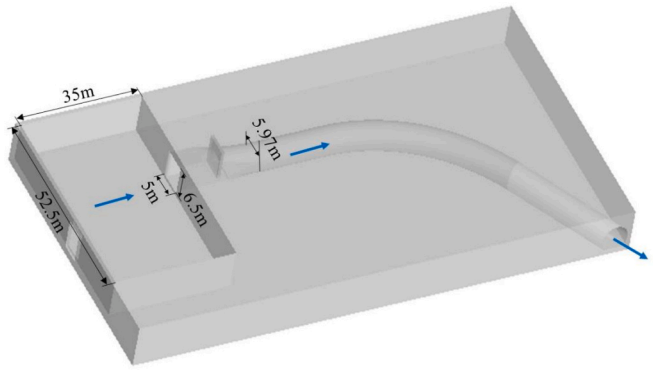


Fig. 1. Three-dimensional model of the south trunk line.

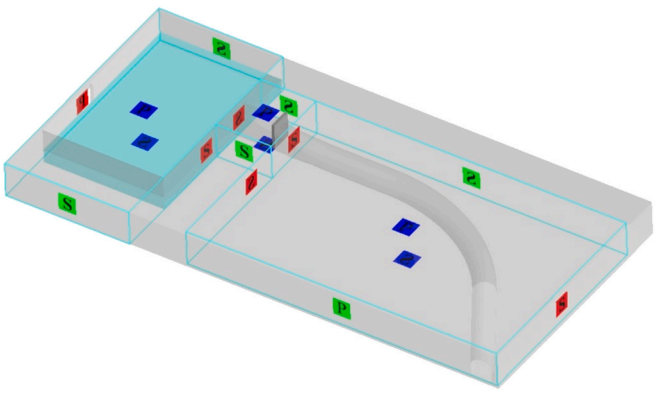


Fig. 3. Boundary condition for submerged flow simulation of sluice gate.

Table 2
Submerged flow numerical simulation condition setting table.

Upstream elevation/m	Downstream water depth/m	Gate opening/m
513.896	2.2	0.11、0.22、0.33、0.44、0.55、
	2.9	0.66、0.77、0.88、0.99、1.10
	3.6	1.21、1.32、1.43、1.54、1.65、
	4.3	1.76、1.87、1.98、2.09、2.20
	5.0	
513.08	1.8	0.09、0.18、0.27、0.36、0.45、
	2.6	0.54、0.63、0.72、0.81、0.90
	3.4	0.99、1.08、1.17、1.26、1.35、
	4.2	1.44、1.53、1.62、1.71、1.80
	5.0	
512.00	1.0	0.05、0.1、0.15、0.2、0.25、0.3、
	2.0	0.35、0.4、0.45、0.5
	3.0	0.55、0.60、0.65、0.7、0.75、0.8、
	4.0	0.85、0.9、0.95、1.0
	5.0	

'hydraulic monomer model experiment of a long-distance water conveyance project' experimental results were compared with the simulation results to complete the verification work.

(1) Velocity

A total of three velocity measurement lines are positioned along the longitudinal axis of the channel, with velocities measured using a Pitot tube. Each line is vertically arranged with three measuring points at the surface, middle and bottom layers, as shown in Fig. 4. Fig. 5 compares the simulated and experimental velocity at the same position along the surface and middle layers. The RMSE and R^2 of the surface and middle layer velocity are 0.053, 0.054 and 0.915, 0.947, respectively, and all simulations maintain a relative error below 3.8 %. It shows that the simulation results are consistent with the physical model test results and meet the requisite engineering precision.

(2) Water surface profile

The water surface elevation of the longitudinal section of the diversion pool is extracted, and Fig. 6 shows the comparison between the simulated and experimental elevation. The water elevation along the pool is equal, and the simulated elevation is between the high and low values of the experiment, which almost coincides with the low value. The number and quality of grids affect the calculation accuracy, resulting in low simulated elevation. However, compared with the low experimental value, MAPE = 0.024 %, RMSE = 0.126, and the relative error range is 0.016 %–0.028 %. The results show that the overall error is small, the calculation result is reliable and meets the accuracy requirements.

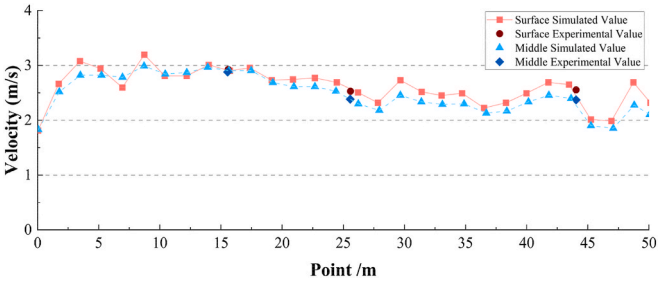


Fig. 5. Comparison of simulated and experimental velocity.

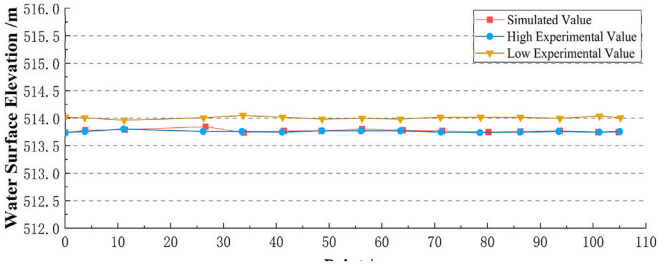


Fig. 6. Comparison of simulation and experimental water surface elevation in the diversion pool.

2.2. Dimensional analysis

The definition sketch of a sluice gate under free and submerged flow conditions is shown in Fig. 7. According to the Bernoulli equation and the continuity equation, the flow calculation formula of the rectangular channel is as follows [24]:

$$Q = C_d \cdot B \cdot e \cdot \sqrt{2gy_1} \tag{11}$$

where C_d is the discharge coefficient; B is the width of the channel; e is gate opening; g is the acceleration of gravity; y_1 is upstream water depth.

For submerged flows, the parameters affecting the C_d can be expressed as:

$$C_d = f_1(Q, y_1, y_3, B, e, \rho, \mu, \sigma, g) \tag{12}$$

where y_3 is downtream water depth; ρ is the fluid density; μ is the dynamic viscosity; σ is the surface tension coefficient.

According to Buckingham-π theorem [25] and dimensional analysis shows that:

$$C_d = f_2\left(Fr, \frac{B}{e}, \frac{y_1}{e}, \frac{y_3}{e}, Re, We\right) \tag{13}$$

where Fr is the Froude number; B/e is the ratio of channel width to gate

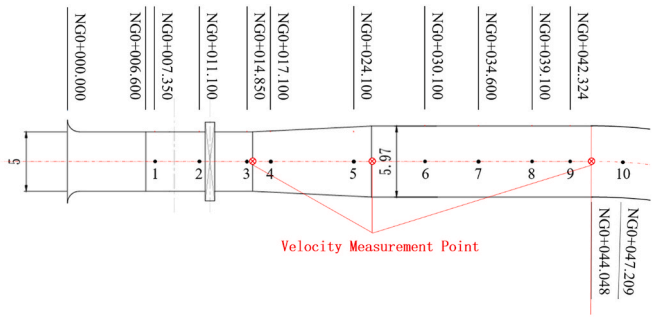


Fig. 4. Arrangement of velocity measuring point in the south trunk line.

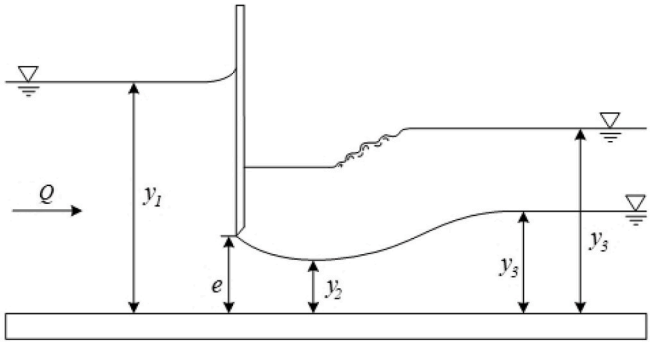


Fig. 7. Schematic of the sluice gate flow.

opening; y_1/e is the ratio of upstream water depth to gate opening; y_3/e is the ratio of downstream water depth to gate opening. Re is the Reynolds number; We is the Weber number.

Given that the Reynolds number across all simulated cases (Re $\approx 10^5$ to 10^8) is well within the turbulent range, and considering the macroscopic scale of the flow, the effects of both viscous forces and surface tension are negligible [26].

Therefore, Re and We can be ignored, Eq. (13) can be simplified as Eq. (14).

$$C_d = f_3\left(Fr, \frac{B}{e}, \frac{y_1}{e}, \frac{y_3}{e}\right) \tag{14}$$

The statistical characteristics of each dimensionless parameter affecting the submerged C_d are reported in Table 3.

2.3. Intelligent model

2.3.1. Random Forests(RF)

Random Forest [27] is a typical representative of ensemble learning method, which is an ensemble algorithm based on decision tree. The algorithm's core is to aggregate multiple learners to balance the algorithm error, and the structure is shown in Fig. 8. A few wrong judgments or predictions generated in the process will be corrected or averaged by other base learners, thus weakening the error and obtaining stronger generalization ability.

RF algorithm constructs a Bagging ensemble, introduces a random selection of features in the training process of the decision tree, and then selects an optimal attribute for partitioning. In the decision tree construction, the 'random' of RF is mainly reflected in random sampling and random features. This ensures the independence between the decision trees, making the voting results unbiased [28]. There are countless decision trees in RF, each of which will vote for the sample and summarize the discriminant results. The model output is determined by the majority voting of the tree [29]. Only when more than half of the decision trees have discriminant errors will the results produce misjudgment, thus significantly improving the model's accuracy.

The RF algorithm includes the following key parameter categories [30].

(1) n_estimators: the number of individual decision trees in the forest. Increasing n_estimators improves model stability and robustness by reducing variance, though at the cost of increased computational burden and training time.
(2) min_samples_leaf: the minimum number of samples required to form a leaf node in each tree. This parameter controls individual tree complexity and preventing over-fitting.
(3) max_features: the number of features randomly selected at each split to consider for best partitioning. By introducing random feature sampling to improves ensemble generalization.
(4) max_depth: the maximum depth allowed for decision tree. It restricts tree complexity and reduces overfitting, especially for large-sample or high-dimensional feature settings.

2.3.2. Firefly-algorithm(FA)

The Firefly Algorithm [31] is a nature-inspired intelligent algorithm constructed by simulating the luminescence characteristics of fireflies in nature. Its inspiration is a brightness attraction of fireflies and bioluminescent communication.

Table 3
Statistical characteristics of each dimensionless parameter.

characteristic parameter	Fr	B/e	y_1/e	y_3/e	C_d
minimum	0.011	2.273	2.075	1.318	0.255
maximum	0.711	100	55	100	0.549
average	0.174	12.376	8.472	9.228	0.390

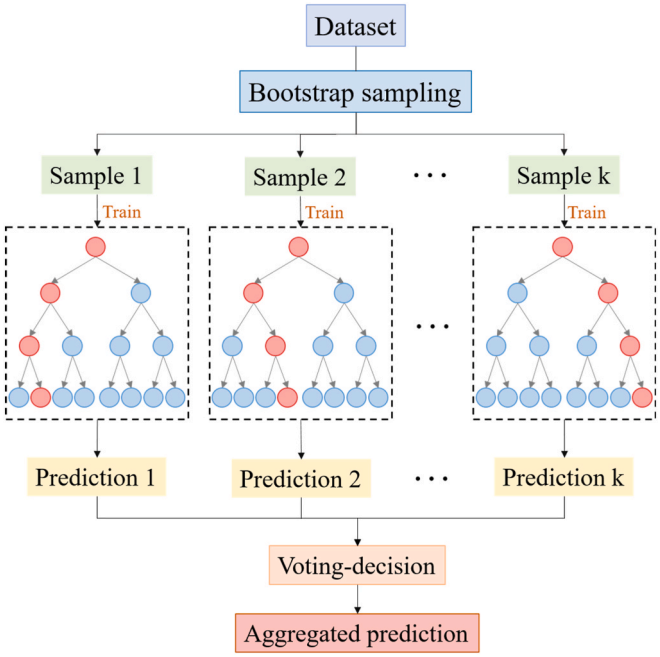


Fig. 8. Diagrammatic representation of Random Forest algorithm.

The light intensity I_i varies according to the. The light intensity represents fitness, the greater the light intensity, the better the position, and the position represents the feasible output:

$$I_i = I_0 e^{-\gamma r^2} \tag{15}$$

I_0 is the maximum brightness, usually 1; γ is the light absorption coefficient, indicating the change of attraction, it varies from 0.1 to 100. r is the distance between fireflies.

The firefly's attractiveness is proportional to the light intensity, which can be calculated by Eq. (16). Firefly i with low brightness will move to j with high brightness:

$$\beta(r) = \beta_0 e^{-\gamma r^2} \tag{16}$$

where β_0 is the initial attractiveness at $r = 0$, and $\beta_0 = 1$;

When the i-th firefly is attracted to another more attractive firefly j, the movement of the position is updated according to Eq. (17):

$$x_i' = x_i + \beta_0 e^{-\gamma r^2}(x_i - x_j) + \alpha \cdot rand() \tag{17}$$

where x_i is the position of low brightness firefly i; *rand (.)* is a random term, N (0, 1); α is the step factor of the perturbation, $\alpha \in (0,1)$.

During the solution process, the firefly moves continuously until the termination condition is satisfied, and the optimal solution is the model output. Compared with other optimization algorithms, FA has a simple operation process and fewer parameters and can find global and local optimal simultaneously [32], which has been widely used.

2.3.3. Firefly algorithm-optimized Random Forest (FA-RF)

In this study, the Firefly Algorithm FA was employed to optimize the hyperparameters of the Random Forest (RF) model. The model inputs include the Froude number Fr, the ratio of channel width to gate opening B/e, the ratio of upstream water depth to gate opening y_1/e and the ratio of downstream water depth to gate opening y_3/e. The discharge coefficient C_d is model output. The parameters of the FA are configured as shown in Table 4, and the RF hyperparameters are provided in Table 5. The objective function was defined as RMSE of the model predictions.

Table 4
Parameters of the FA.

Model	Population size	Max iterations	γ	β_0	α
FA	20	50	1.0	1.0	0.25

Table 5
Search ranges of RF hyperparameters.

Model	n_estimators	min_samples_leaf	max_features	max_depth
RF	[50, 500]	[1,20]	[0.1, 1.0]	[5,20]

2.4. Model evaluation indicators

In order to comprehensively analyze the robustness, data consistency and stability of the model, six statistical indicators, including correlation coefficient (R), mean absolute percentage error (MAPE), Nash-efficiency coefficient (NSE), root mean square error (RMSE), Wilmot consistency index (WIA) and objective function standard (OBJ), were selected to evaluate the performance of the model from multiple aspects comprehensively. Each indicator is defined as follows.

$$R = \frac{\sum_{i=1}^{N}(O_i - O_a)(P_i - P_a)}{\sum_{i=1}^{N}(O_i - O_a)\sum_{i=1}^{N}(P_i - P_a)} \tag{18}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{O_i - P_i}{O_i}\right| \times 100\% \tag{19}$$

$$NSE = 1 - \frac{\sum_{i=1}^{N}(O_i - P_i)^2}{\sum_{i=1}^{N}(O_i - \overline{O_i})^2} \tag{20}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(O_i - P_i)^2}{N}} \tag{21}$$

$$WIA = 1 - \frac{\sum_{i=1}^{N}(P_i - O_i)^2}{\sum_{i=1}^{N}(|P_i - O_a| + |O_i - O_a|)^2} \tag{22}$$

$$OBJ = \left(\frac{N_a - N_e}{N}\right) \times \left(\frac{\mathrm{MAE_a}}{R_a^2}\right) + \left(\frac{2N_e}{N}\right) \times \left(\frac{\mathrm{MAE_e}}{R_e^2}\right) \tag{23}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|O_i - P_i| \tag{24}$$

where O_i is the discharge coefficient's experimental value, O_a is the average value of the measured discharge coefficient, P_i is the discharge coefficient's predicted value, and P_a is the average value of the predicted discharge coefficient; N is the total number of samples in the dataset; N_a is the number of training samples, N_e is the number of test samples; $\mathrm{MAE_a}$ is the mean absolute error of the training datasets; $\mathrm{MAE_e}$ is the mean absolute error of the testing datasets; R_a^2 is the determination coefficient of the training datasets. R_e^2 is the determination coefficient of the testing datasets.

2.5. Uncertainty analysis

Model uncertainty analysis can effectively demonstrate the predictive capacity of numerical models and verify the data relationship. The formula for calculating the mean error ($\overline{me}$) of the model is shown in Eq. (25). The model is considered underestimated and overestimated if the values of mean error $\overline{me}$ are negative and positive, respectively [33].

$$\overline{me} = \frac{\sum_{i=1}^{N}\overline{me_i}}{N} \tag{25}$$

The standard error (S_{me}) is:

$$S_{me} = \sqrt{\frac{\sum_{i=1}^{N}(me_i - \overline{me})^2}{N-1}} \tag{26}$$

$$me_i = P_i - O_i \tag{27}$$

where me_i is the prediction error of a single sample.

Create confidence bound by using parameters $\overline{me}$ and S_{me}. The 95 % confidence interval (95 %CI) suggests that we can be 95 % confident that the true mean error of our model's predictions falls within this bounded range, and the smaller the interval, the better the performance [34].

$$95\%\mathrm{CI} = \pm 1.96 * S_{me} \tag{28}$$

where CI is the confidence interval of a sample.

2.6. Sobol's sensitivity analysis

The Sobol method [35] is a sensitivity analysis approach based on variance decomposition, which can handle highly nonlinear and non-monotone functions and models. By calculating the variance decomposition of each parameter and its interaction, the influence of the model inputs on the output can be analyzed. The Sobol method is mainly divided into two stages: selecting the sample points of the initial input parameters and substituting the selected sample points into the model [36]. The core is decomposing the objective function into an incremental sum of dimensions [37].

Assuming the input variable $X = (x_1 < x_2 < \ldots < x_n)$ are independent and each term is chosen with zero average and is square integrable, the objective function $f(x)$ can be expressed as:

$$f(x) = f_0 + \sum_{i=1}^{k} f_i(x_i) + \sum_{1<i<j<n} f_{ij}(x_i, x_j) + \ldots + f_{1,2,\ldots,n}(x_1, x_2, \ldots, x_n) \tag{29}$$

where f_0 is a constant in the objective function; i, j are the variable number; $f(x_i)$ is a function of x_i; $f(x_i, x_j)$ is a function of x_i and x_j.

The deviation of each order is:

$$D_{i_1,i_2,L,i_s} = \int_0^1 L \int_0^1 f_{i_1,i_2,L,i_s}^2(x_{i_1}, x_{i_2}, \mathrm{K}, x_{i_s})\mathrm{d}x_{i_1}\mathrm{d}x_{i_2}\mathrm{L}\,\mathrm{d}x_{i_s} \tag{30}$$

where $1 \le i_1 < i_2 < \cdots < i_s \le n (1 \le s \le n)$

The normalized total variance D can be evaluated by Eq. (31):

$$D = \sum_{i=1}^{n} D_i + \sum_{1<i<j<k} D_{ij} + \ldots + D_{1,2,\ldots,n} \tag{31}$$

The Sobol sensitivity indices is the ratio of partial variance to total variance. The first-order sensitivity reflects the sensitivity of a single input parameter to the output, which can be expressed as:

$$S_i = \frac{D_i}{D} \tag{32}$$

The global sensitivity demonstrates the sensitivity to the output after the input parameters interact, which can be expressed as:

$$S_{Ti} = 1 - \frac{D_{\sim i}}{D} \tag{33}$$

where D is the total variance of the output; D_i is the first-order variance of x_i. $D_{\sim i}$ is the variance of the interaction of other parameters except for x_i.

3. Results and discussion

3.1. Hydraulic characteristic analysis

3.1.1. Flow pattern

To explore the three-dimensional flow field structure of the submerged flow of the sluice gate, this study selected the streamline of 'upstream water elevation 513.08 m, $e = 0.18$ m, Y-axis direction Y = 0 m' for analysis, as shown in Fig. 9. Due to the small opening of the gate, when $y_3 = 1.8$ m, the downstream flow has enormous specific energy. The flow rolls upstream, and the contraction section is drowned, forming a submerged hydraulic jump on the water surface. When $y_3 = 2.6$ m and 3.4 m, the submergence degree increases, and the submerged hydraulic jump position shifts to underwater. The hydraulic jump length decreases with the rise of y_3, and there is a tendency of backflow upstream. Because the dynamic energy of the downstream flow increases, the disturbance to the upstream flow increases, thus shortening the range of submerged hydraulic jumps. When $y_3 = 4.3$ m and 5.0 m, the downstream water depth exceeds the upstream water depth, the energy cannot be counterbalanced, and the submerged hydraulic jump behind the gate disappears. Only a tiny part of the flow can pass through the gate, most of the remaining water flows in the opposite direction under the obstruction of the gate, so there is backflow upstream.

3.1.2. Flow velocity

In order to explore the internal variation law of the submerged flow of the sluice gate, the upstream water elevation of 513.08 m and $e = 0.18$ m is selected for analysis, as shown in Fig. 10. When $y_3 = 1.8$ m, the flow velocity under and behind the gate is large, and the rest is negligible. The maximum velocity under the gate is 6.8 m/s, decreasing vertically from the bottom to the top. When $y_3 = 2.6$ m, the high-velocity area is still downstream, but the overall velocity decreases, and the velocity near the hydraulic jump is slightly larger than the surrounding. When $y_3 = 3.4$ m, the upstream and downstream water depths are almost equal. The maximum velocity is 1.0 m/s, and the high-velocity area moves from downstream to upstream. When $y_3 = 4.2$ m and 5 m, the increase of y_3 decreases the flow velocity.

3.1.3. Hydraulic jump

Hydraulic jump is the process of transition from supercritical flow to subcritical flow in a free gravitational flow with violent turbulence and energy dissipation [38]. It occurs widely in most hydraulic structures [1]. For examining the phenomenon and variation law of hydraulic jump, the side view with upstream water elevation of 513.08 m and $y_3 = 1.8$ m is adopted for analysis. As shown in Fig. 11, when $e \geq 0.54$ m, the conjugate water depth of the contraction section exceeds the downstream water depth. The turbulence through the gate will continue to flow downstream for a certain distance, and a part of the dynamic energy is consumed by friction during the process. The flow velocity decreases, and the downstream water depth increases gradually until it equals the conjugate water depth, forming a free hydraulic jump. At this time, the downstream flow has no effect on the upstream. As the gate opening decreases, the hydraulic jump position gradually approaches the gate. When $e = 0.18$ m, the tailwater depth of the flow from a sluice gate is greater than the subcritical sequent depth, the gate outlet is drowned and the jump becomes submerged [39]. The submerged hydraulic jump has a complicated flow structure, and the discharge will decrease with the increase of downstream water depth.

3.2. Prediction model analysis

According to the dimensional analysis, the dimensionless parameters affecting the submerged discharge coefficient C_d of the sluice gate include the Froude number Fr, the ratio of channel width to gate opening

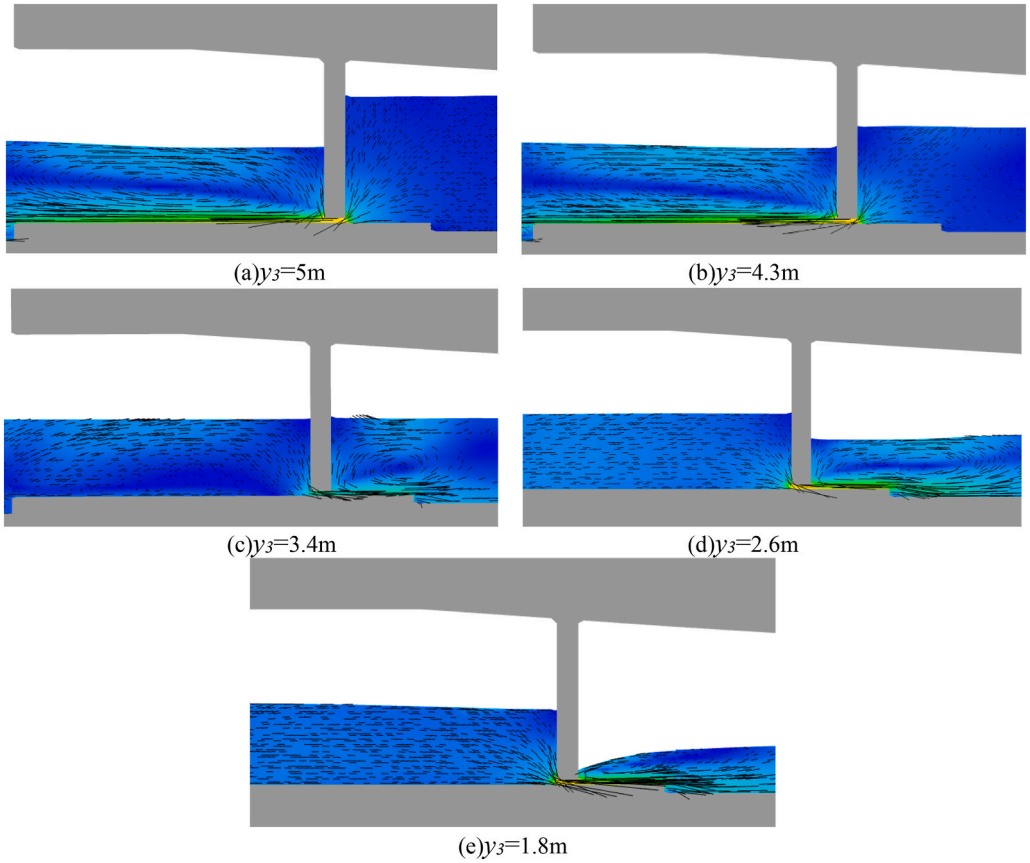


Fig. 9. The velocity vector of submerged flow with different downstream water depths.

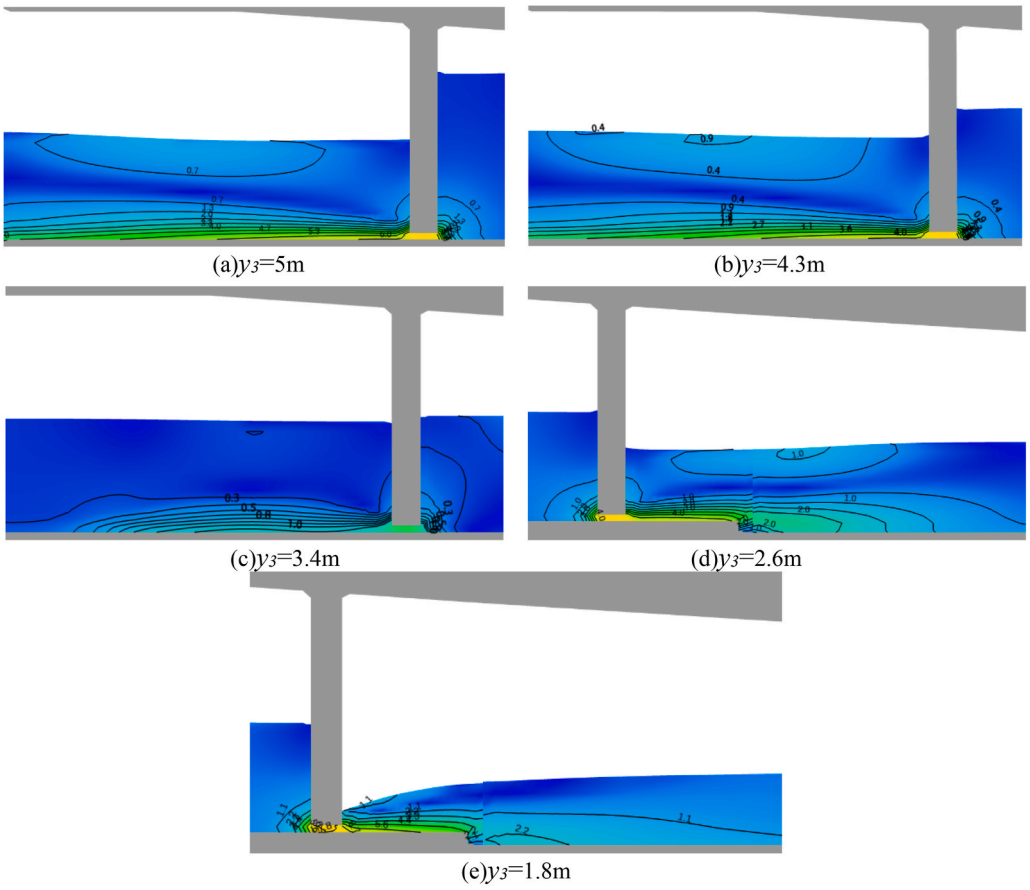


Fig. 10. The velocity contour of submerged flow with different downstream water depths.

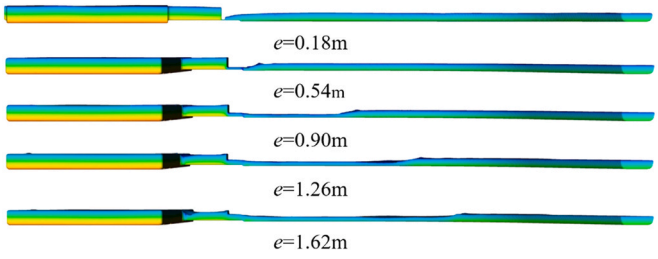


Fig. 11. Side view of submerged flow with different gate openings.

B/e, the ratio of upstream water depth to gate opening y_1/e and the ratio of downstream water depth to gate opening y_3/e. Therefore, the above dimensionless parameters are model inputs, and the C_d is model output.

The performance indexes of RF and FA-RF models in the training and testing phase are shown in Table 6. It can be seen that both models have acceptable accuracy, but the performance of FA-RF is better than RF. The MAPE and RMSE of FA-RF are reduced by 20.9 % and 24.3 %, respectively, compared with RF in the testing phase, which shows that FA-RF has more significant error reduction and greater performance improvement. The objective function(OBJ) values of RF and FA-RF are 0.024 and 0.018, respectively, indicating that the prediction results of FA-RF have the strongest correlation with the experimental results and the sample parameters have the most substantial credibility.

Table 6
Model performance evaluation indices of submerged flow.

phase	model	MAPE(%)	RMSE	R	NSE	WIA	OBJ
train	RF	3.3977	0.0168	0.974	0.941	0.983	0.024
	FA-RF	3.2497	0.015	0.983	0.953	0.986	0.018
test	RF	4.372	0.037	0.945	0.889	0.969	0.024
	FA-RF	3.455	0.028	0.971	0.93	0.98	0.018

Fig. 12 presents the scatter plots of predicted and simulated C_d of the two models in the training and testing phases, and both models have superior performance overall. In the training phase, the predicted data points of FA-RF are closer to the 1:1 fitting line than RF. However, when $0.25<C_d<0.3$, the predicted C_d of the two models are greater than the simulated values, especially RF. When $0.4<C_d<0.55$, the predicted C_d is lower than the simulated C_d. The data distribution of RF and FA-RF models in the testing phase is more uniform than in the training phase. The data points are almost symmetrically distributed about the fitting line, and there is no apparent trend of overestimation or underestimation. The R of FA-RF is improved compared with RF, and the optimized model has more minor errors and better generalization ability, showing a better prediction effect.

The relative deviation (RD) of the predicted C_d relative to the simulated C_d is shown in Fig. 13. In the training phase, the data distribution shows a trend of tilting from the upper left to the lower right. When $C_d<0.3$, the predicted C_d is greater than the simulated C_d, but it's opposite when $C_d>0.4$. In the testing phase, the RD range of RF and FA-RF is [−0.068,0.056] and [−0.037,0.048], respectively. When $C_d<0.35$, the prediction results of RF are on the high side, while FA-RF doesn't have the same trend. In short, FA-RF has a smaller RD range, evenly distributed around 0, indicating higher accuracy and model stability.

Fig. 14 is the error density diagram in the testing phase, and the error is mainly distributed at both ends of C_d. The error of RF is more dispersed, more prominent and asymmetric. FA-RF model's error is symmetrical about the 0.01 line. Although there are also overestimations, the range is smaller, and the distribution is more uniform

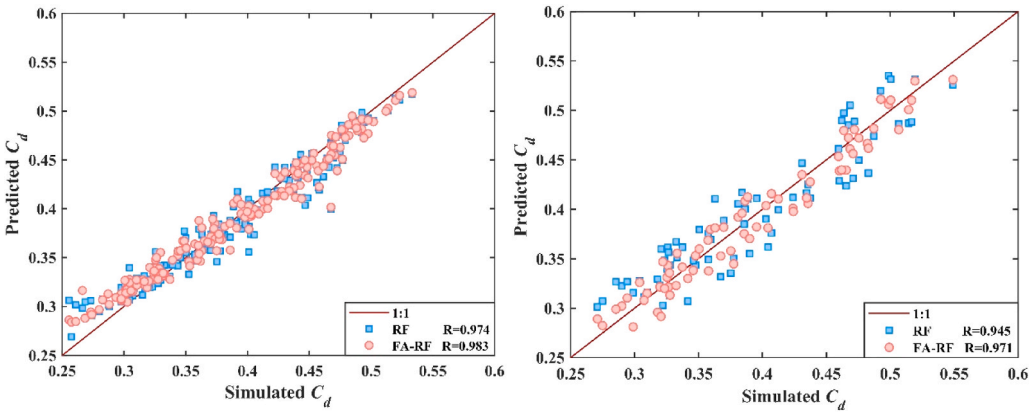


Fig. 12. Scatter plot of predicted and simulated C_d.

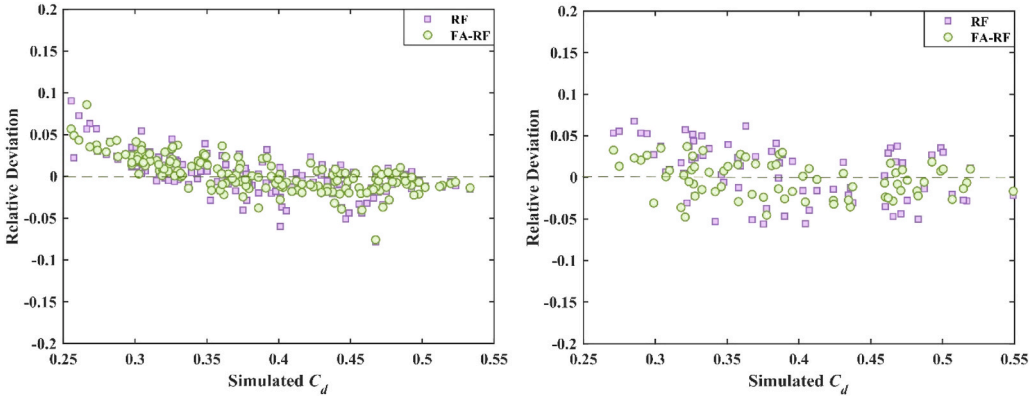


Fig. 13. Relative deviation of different prediction models.

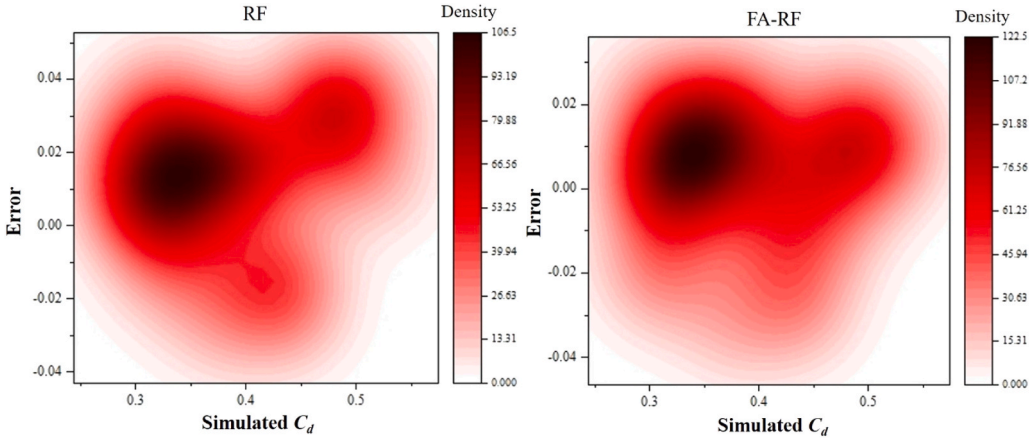


Fig. 14. Error density in the testing phase.

and concentrated.

Fig. 15 shows the Taylor diagram for two models in the testing stage. The three performance indicators of RMSE, R and SD are integrated into one point; the one closest to the observation point represents the best performance [40]. The distance from the FA-RF model to the observation point is shorter than that of RF, indicating that its comprehensive prediction performance is better. In summary, the Firefly algorithm optimization improves the model's comprehensive performance.

The uncertainty analysis results of the testing phase are shown in Table 7. The mean error $\overline{me}$ of RF and FA-RF is 0.00330 and −0.00247, respectively, with a slight overestimation and underestimation. The model can be considered stable since the absolute value of $\overline{me}$ is less than 0.01. For the FA-RF model, the standard deviation $S_{me} = 0.00186$. The 95 % confidence interval (95 % CI) for the mean error is [−0.00612, 0.00118], with a total width of 0.00729. The S_{me} of FA-RF is 38.7 % lower than RF, indicating that FA-RF has more accurate control of the prediction results, significant data stability, robust model performance, and high reliability of sample parameters.

In summary, the FA-RF model is the optimal intelligent prediction model for the submerged discharge coefficient of the sluice gate.

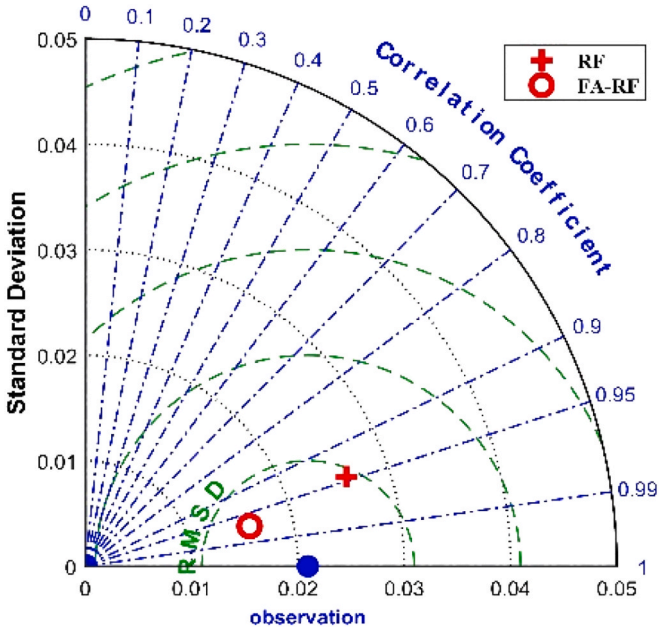


Fig. 15. Taylor diagram of different prediction models.

Table 7
Uncertainty analysis results of submerged flow in the testing phase.

Model	$\overline{me}$	S_{me}	Band width	95 %CI	
RF	0.00330	0.00304	0.01192	−0.00266	0.00926
FA-RF	−0.00247	0.00186	0.00729	−0.00612	0.00118

3.3. Sensitivity analysis of hydraulic parameters

The appropriate number of Sobol samples is significant for sensitivity analysis. The first-order sensitivity coefficient S_i and the global sensitivity coefficient S_{Ti} change with the sampling times, as shown in Fig. 16. When the sampling time is small, the sensitivity coefficient fluctuates wildly. The initial values of *Fr* and y_3/e are relatively high and low, respectively, and the fluctuation of B/e and y_1/e is slight. The sensitivity coefficient gradually reaches a stable value with the increase of sampling times. Therefore, the final adopted number of Sobol Samples is 200,000.

To further explore the influence of each input parameter on submerged C_d, the Sobol method was introduced to quantify the sensitivity of hydraulic parameters. From Fig. 17, the S_i of the Froude number *Fr*, ratio of channel width to gate opening B/e, ratio of upstream water depth to gate opening y_1/e and ratio of downstream water depth to gate opening y_3/e are 0.5428, 0.0105, 0.097 and 0.3562. Among them, B/e and y_1/e account for the smallest proportion, only about 1 %, *Fr* and y_3/e account for 59 % and 38.8 % respectively. It shows that only considering a single parameter, *Fr* and y_3/e are important parameters affecting submerged C_d. After parameter interaction, the S_{Ti} of *Fr*, B/D, y_1/e and y_3/e are 0.6004, 0.0378, 0.0353 and 0.412, respectively. The greatest impact on C_d is still *Fr*, accounting for 55 %, followed by y_3/e, accounting for 37.9 %. These two parameters have the most significant impact on submerged C_d. Therefore, in evaluating the submerged discharge capacity of the sluice gate, *Fr* and y_1/e should be considered emphatically.

Fig. 18 shows the relationship between the core parameters *Fr* and y_3/e with C_d. Fig. 18(a) shows that C_d increases with the increase of *Fr*. When *Fr* is fixed, the larger the y_3, the smaller the C_d. With the increase of y_3, the increasing trend of C_d becomes larger, and the range decreases. Fig. 18(b) shows that C_d decreases with the increase of y_3/e, showing a curve extending from the upper left to the lower right. When $y_3/e < 5$, C_d decreased sharply; When $y_3/e = 5$, the inflection point of the curve appears, the curvature decreases, and then gradually tends to be gentle. When the *e* is fixed, the greater the y_3, the hydraulic jump submerged the contraction section, the greater the interference of downstream flow to

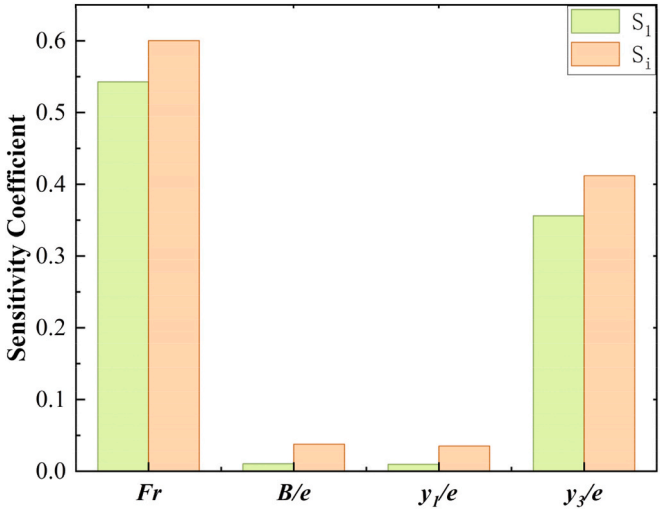
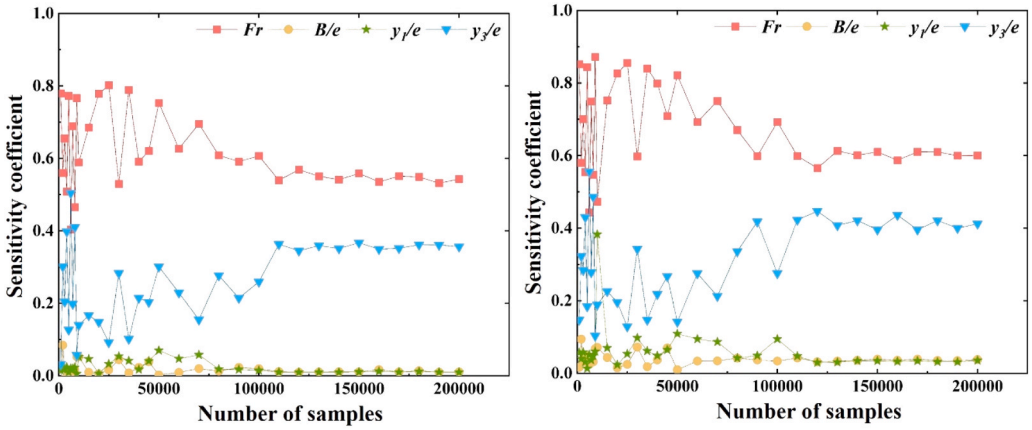
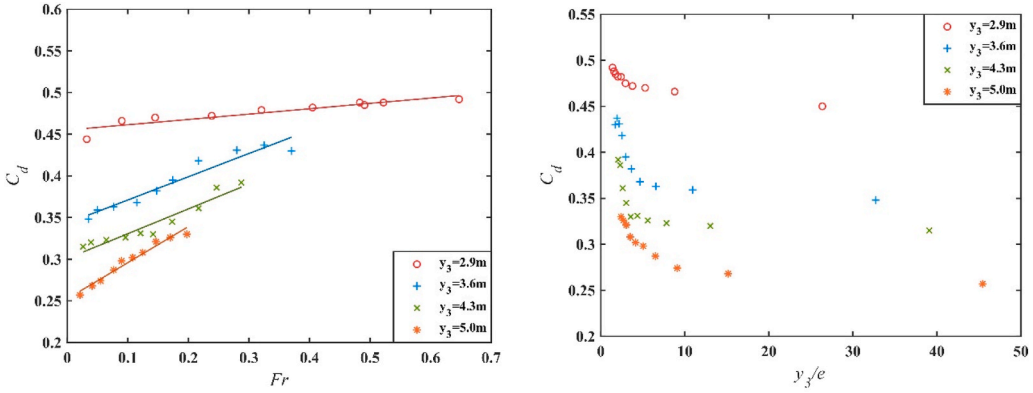


Fig. 17. Sensitivity coefficient of hydraulic parameters.



(a) The first-order sensitivity coefficient S_i (b) The global sensitivity coefficient S_{Ti}

Fig. 16. The change of sensitivity coefficient with number of Sobol samples.



(a)The variation of C_d with Fr (b)The variation of C_d with y_3/e

Fig. 18. The relationship between dimensionless parameters and C_d.

the upstream flow, and the flow through the gate decreases. Therefore, the increase of y_3 leads to a greater downward trend of C_d, a smaller range and a more dispersed data distribution.

4. Conclusion

As a key technology in flood warning and water resources optimization, submerged flow affects urban waterlogging control and water conservancy project safety management, and its calculation and analysis are very important. This study focuses on the discharge coefficient prediction and sensitivity analysis of submerged flow in sluice gates. The submerged flow was numerically simulated, and its hydraulic characteristics were analyzed. Two intelligent models, RF and FA-RF, were developed to predict the submerged discharge coefficient C_d. Sobol sensitivity analysis was employed to quantitatively assess the dimensionless parameters influencing, identify core hydraulic parameters, explore parameter interactions and variation patterns. The conclusions are as follows.

(1) Under fixed upstream and downstream water depths, the gate opening e does not affect the submergence degree; Increasing downstream water depth transitions the flow from free to submerged hydraulic jump, causing upstream backflow. The high-velocity area shifts upstream, and the gate discharge gradually decreases.
(2) In the testing phase, the FA-RF model has MAPE = 3.455 %, RMSE = 0.028, R = 0.971, NSE = 0.93, WIA = 0.98, which is the optimal intelligent prediction model for predicting the submerged flow of the sluice gate. Optimized by the firefly algorithm, The FA-RF model exhibits stable performance and strong robustness. It effectively predicts submerged flow, aiding in evaluating flood and submerged area changes, and provides a reliable reference for water resource scheduling and emergency flood management.
(3) The Sobol sensitivity analysis shows that the first-order sensitivity coefficient S_i and global sensitivity coefficient S_{Ti} of the upstream Froude number Fr, the ratio of gate width to gate opening B/e, the ratio of upstream water depth to gate opening y_1/e, and the ratio of downstream water depth to gate opening y_3/e are 0.5428, 0.0105, 0.097, 0.3562 and 0.6004, 0.0378, 0.0353, 0.412, respectively. Whether considering single parameter or parameter interaction, Fr and y_3/e are the core parameters affecting submerged C_d. In addition, C_d increases with Fr, the greater y_3, the greater the growth trend of C_d, and within a smaller the range. C_d decreases with y_3/e, and when e is fixed, a larger y_3 leads to a sharper decline in C_d.

CRediT authorship contribution statement

Dingye Cao: Writing – original draft, Visualization, Resources, Conceptualization. **Guodong Li:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Shanshan Li:** Visualization, Validation, Funding acquisition.

Ethical approval

There are no relevant waivers or approvals.

Consent to participate

Authors give their permission.

Consent for publication

Authors allow publication if the research is accepted.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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